

EX 13:

Implement the Car Price Prediction Model using Python

PROGRAM :

```
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Car dataset
data = {
    'Year': [2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024],
    'Kilometers': [90000, 80000, 70000, 60000, 50000,
                   40000, 30000, 20000, 10000, 5000],
    'Engine': [1200, 1200, 1500, 1500, 1600,
               1600, 1800, 1800, 2000, 2000],
    'Price': [300000, 350000, 400000, 450000, 500000,
              600000, 700000, 800000, 900000, 1000000]
}

df = pd.DataFrame(data)

# Input and output
X = df[['Year', 'Kilometers', 'Engine']]
y = df['Price']

# Split data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
```

```
)
```

```
# Create and train model
```

```
model = LinearRegression()
```

```
model.fit(X_train, y_train)
```

```
# Prediction
```

```
y_pred = model.predict(X_test)
```

```
print("Actual Prices:")
```

```
print(y_test.values)
```

```
print("\nPredicted Prices:")
```

```
print(y_pred.round(2))
```

```
print("\nMean Squared Error:")
```

```
print(mean_squared_error(y_test, y_pred))
```

```
print("\nR2 Score:")
```

```
print(r2_score(y_test, y_pred))
```

```
# Predict price of a new car
```

```
new_car = pd.DataFrame(
```

```
    [[2022, 25000, 1800]],
```

```
    columns=['Year', 'Kilometers', 'Engine']
```

```
)
```

```
price = model.predict(new_car)
```

```
print("\nNew Car Details:")
```

```
print("Year: 2022")
```

```
print("Kilometers: 25000")
```

```
print("Engine: 1800 CC")
```

```
print("\nPredicted Car Price:")
```

```
print(round(price[0], 2))
```

OUTPUT :

```
Actual Prices:
[900000 350000]

Predicted Prices:
[818244.8 352078.52]

Mean Squared Error:
3344116188.1553

R2 Score:
0.9557802818095167

New Car Details:
Year: 2022
Kilometers: 25000
Engine: 1800 CC

Predicted Car Price:
862586.61
```